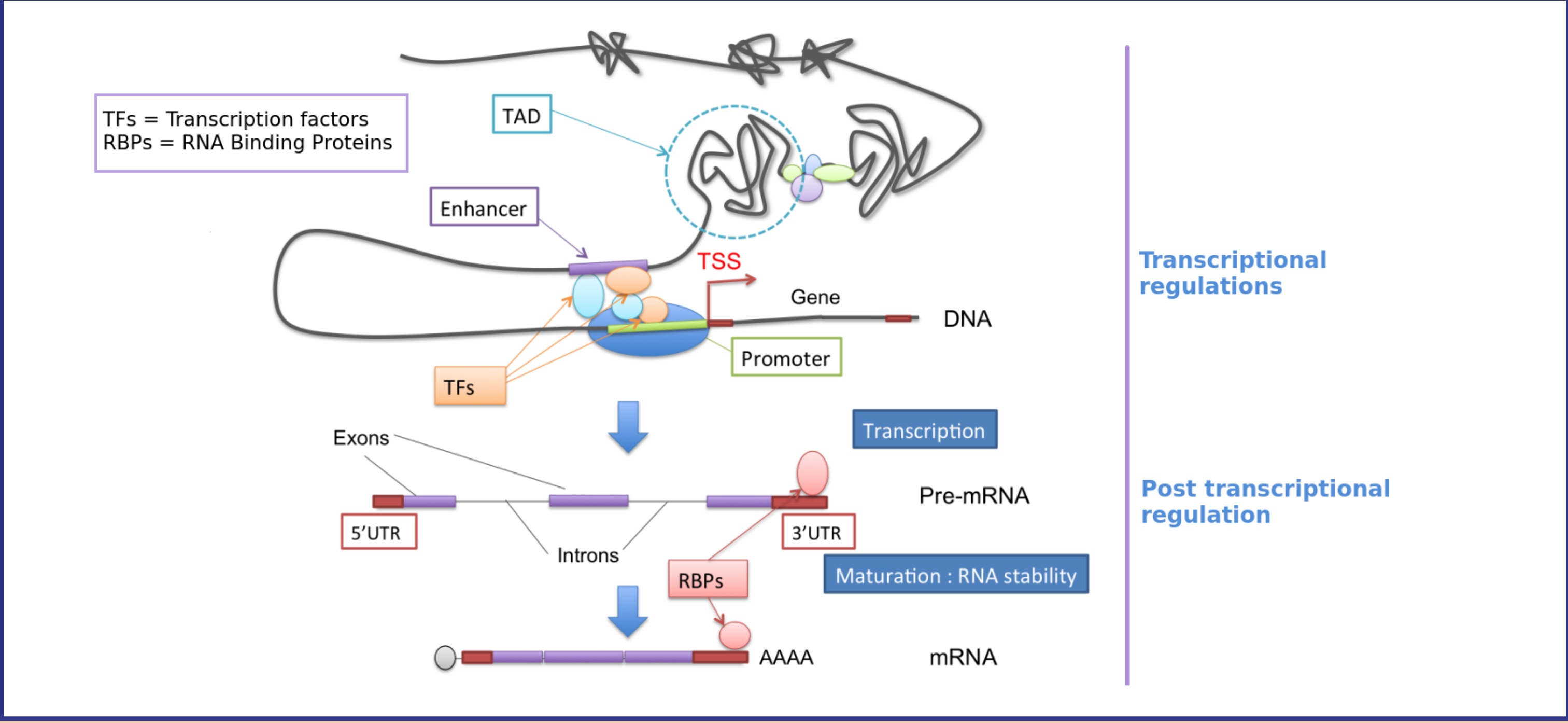


Abstract

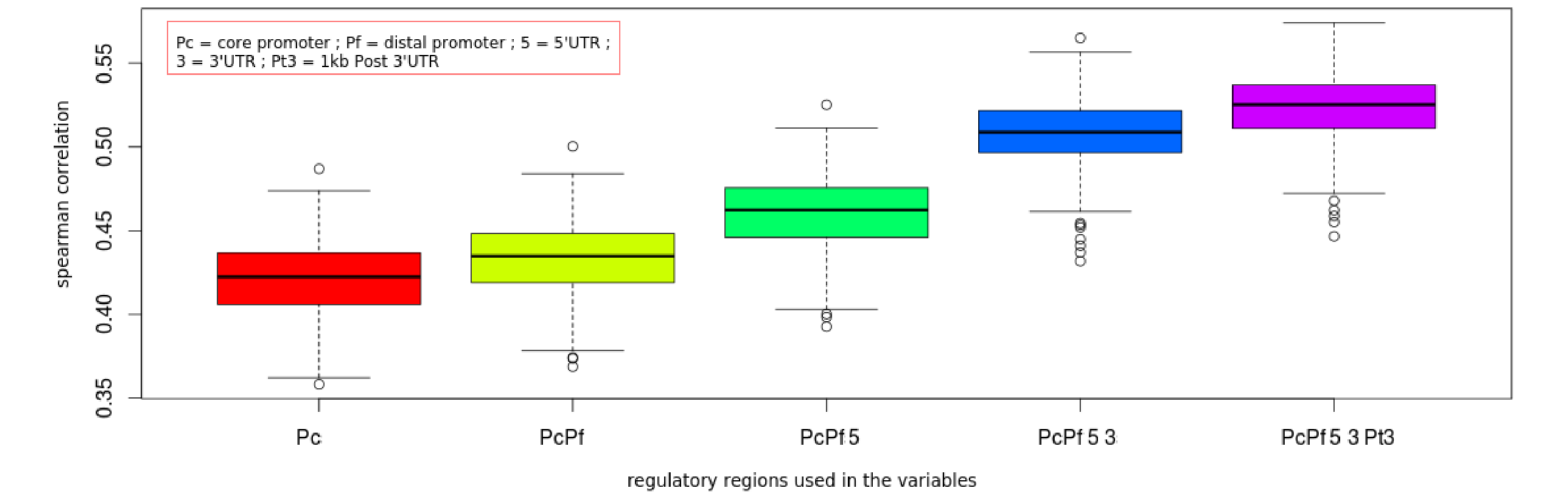
Gene expression is tightly controlled to ensure a wide variety of cell types and functions. Our project aims to model the link between RNA expression and DNA features in regulatory regions (typically, presence/absence of transcription factor motifs in promoters). Penalized linear regression (LASSO) is suitable for this problem, that requires selecting variables in high dimensional data. With this approach, we were able to model gene expression in several cancers. But the inferred model is not equally efficient for all genes. Our perspective is now to identify clusters of coregulated genes and train a specific regression model for each cluster with the right features. **Keywords : regulatory network inference - cis-regulatory motifs - regulatory sequence - cancer**

Regulation of gene expression



Other regulatory regions (nt rates)

Correlation coefficient between measured expression and predicted expression for all cancers, adding other regulatory regions



Gene expression data & DNA features

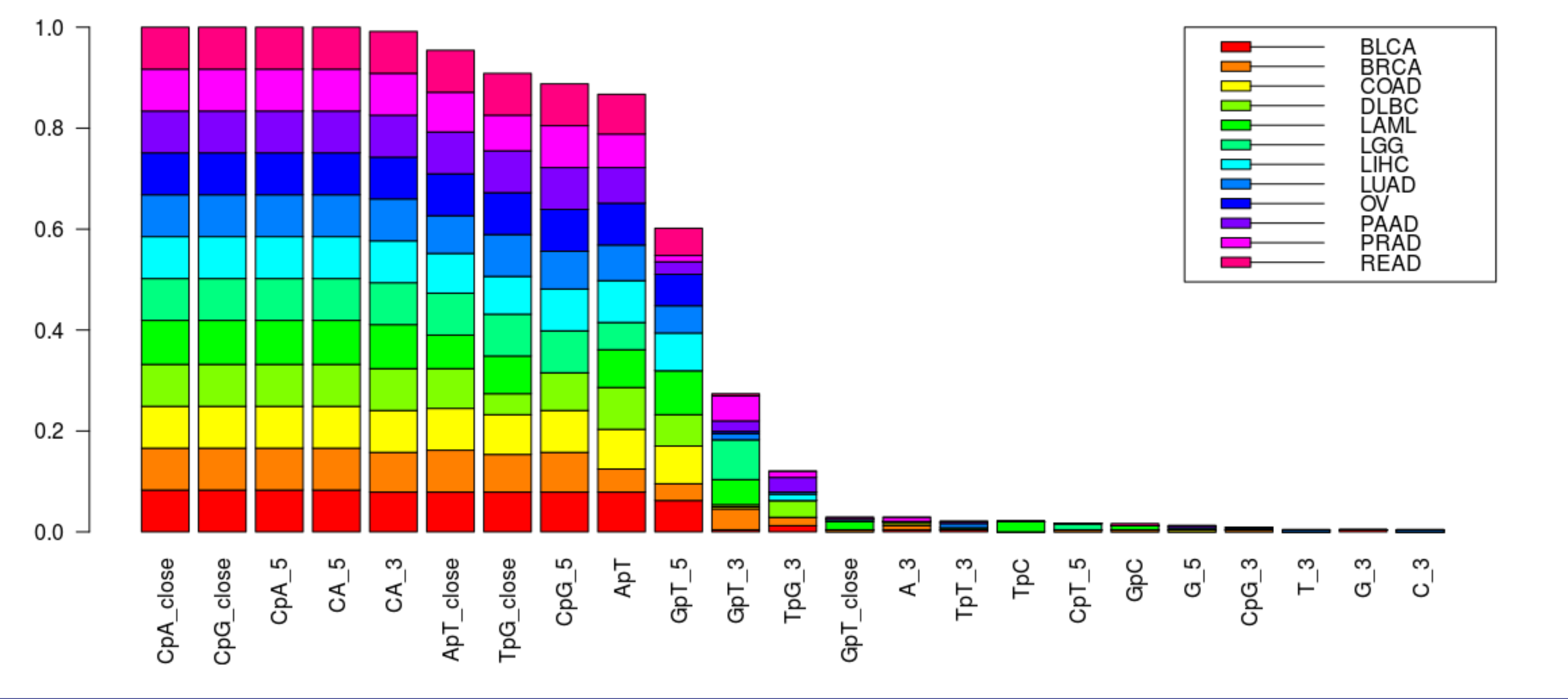
	Predicted variable			Predictive variables					
Genes	Pat_1	..	Pat_{241}	Mo_1	..	Mo_{471}	Nuc_1	..	Nuc_{26}
Ge_1	3.87	..	5.27	0.87	..	0.89	0.06	..	0.22
Ge_2	6.12	..	5.42	0.86	..	0.91	0.27	..	0.3
..		..		..		..		..	
..		..		..		..		..	
Ge_n	5.75	..	6.1	0.89	..	0.91	0.24	..	0.3

Gene expression in 12 different cancers (breast cancer, leukemia...):
RNA-seq (TCGA) → read counts for 241 patients

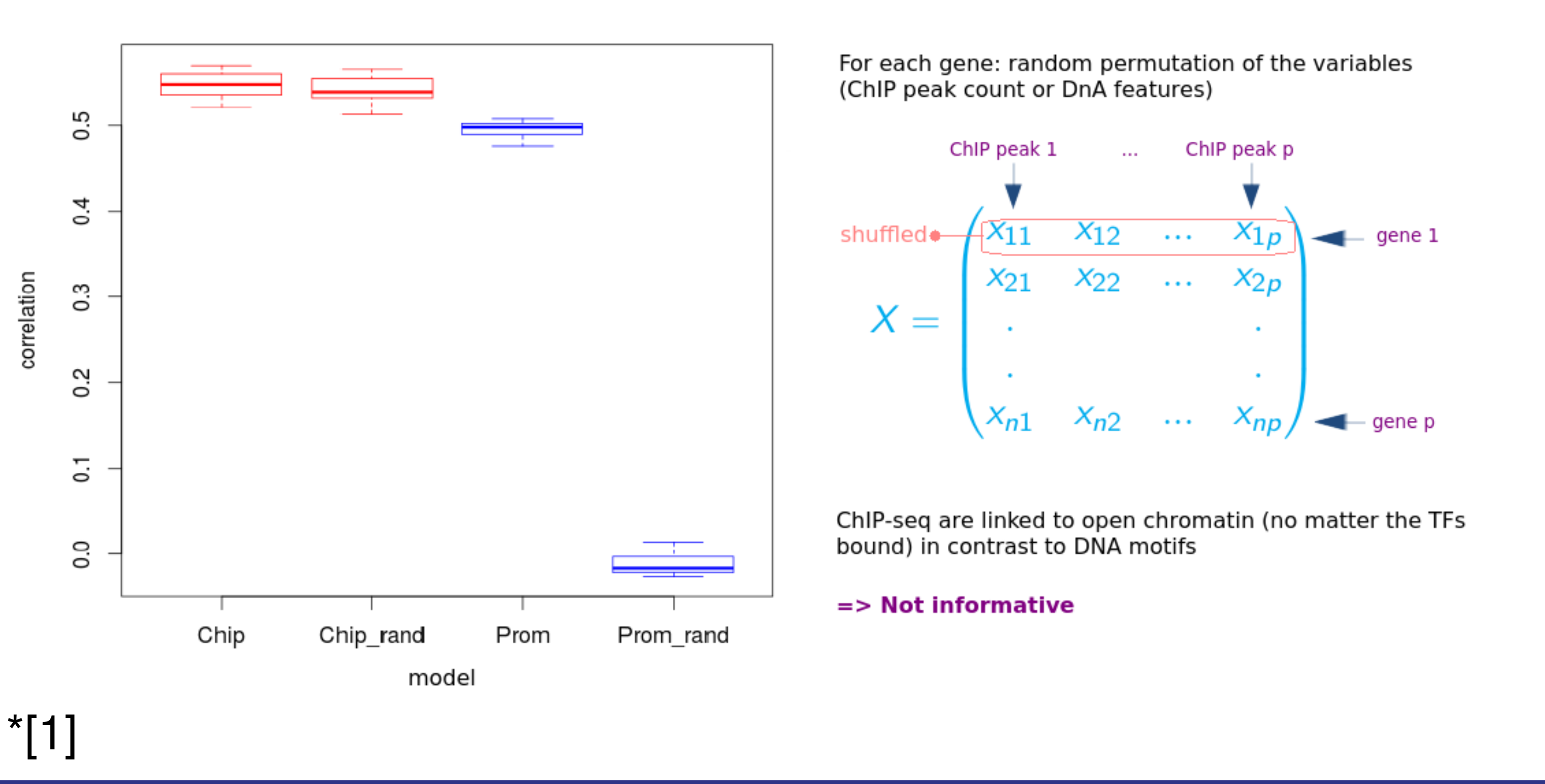
Motif scores: PWM matrix (JASPAR 2016) → Probability of presence of the motif

Nucleotide & dinucleotide rates in regulatory regions (Promoter, 5'UTR, 3'UTR...)

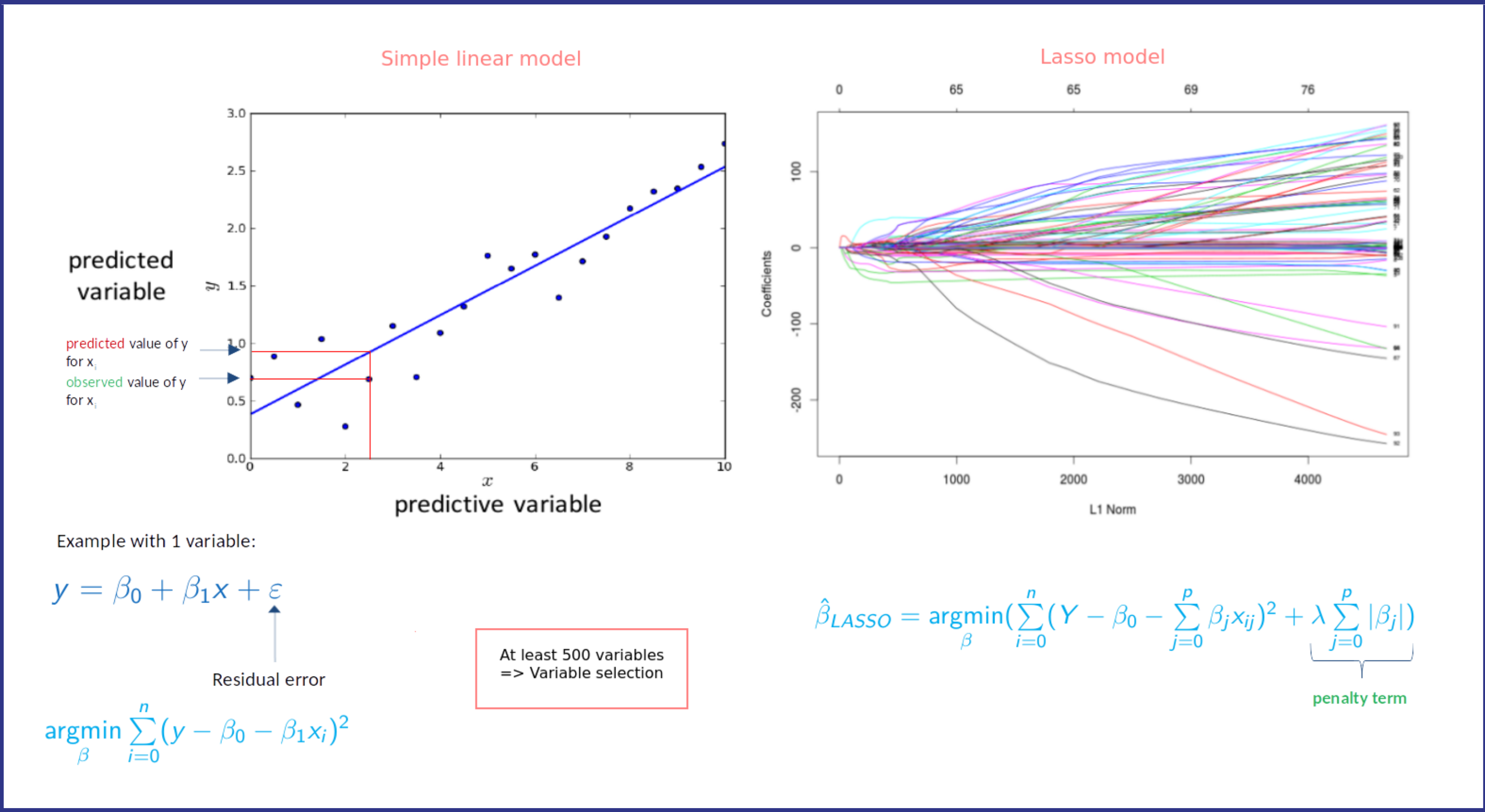
Selected variables



Model comparison: ChIP-seq* VS DNA features

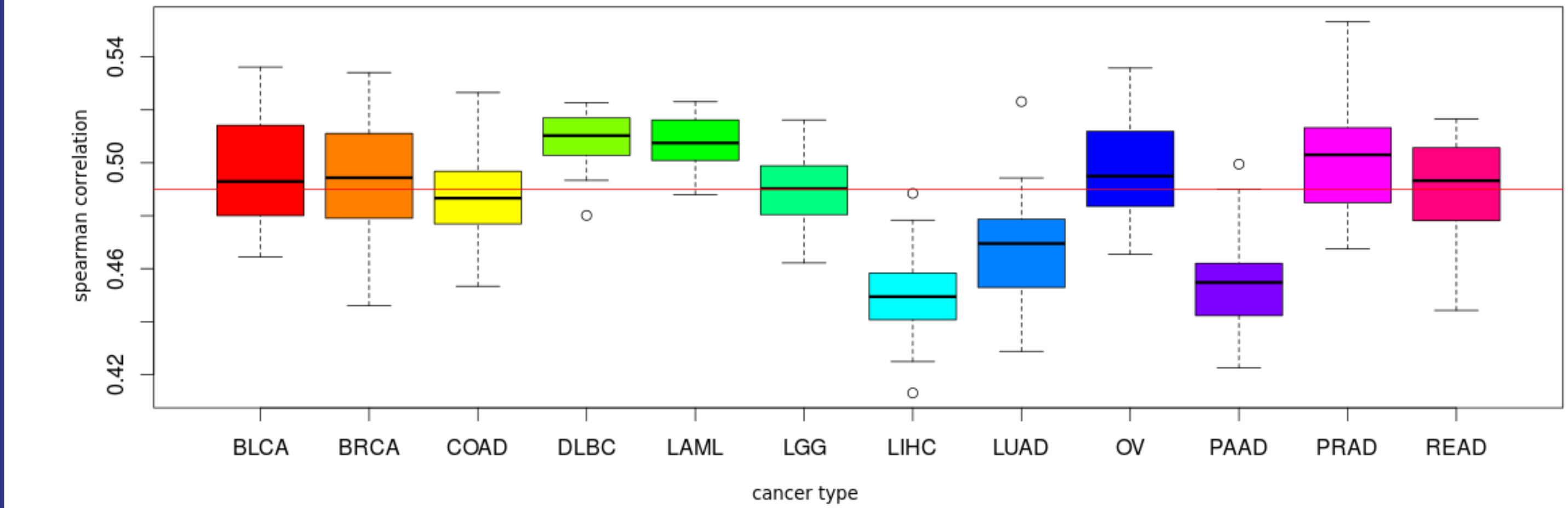


Linear model [2]



Results with the promoter (nt rates & motifs)

Correlation coefficient between measured expression and predicted expression
Boxplot set on 20 tumors for each cancer type



Perspectives

- Identify groups of genes → fit a model for each group
- Biological interpretation:
 - TADs (Topologically Associated Domains) → nt composition
 - Well predicted genes VS poorly predicted genes
 - Some of particular TFs known to have a role in some cancers
- Generalize to the non coding RNA: miRNA and long noncoding RNA

References

[1] Yue Li, Minggao Liang, and Zhaolei Zhang. Regression analysis of combined gene expression regulation in acute myeloid leukemia. *PLoS Comput Biol*, 10(10):e1003908, 2014.

[2] Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society. Series B (Methodological)*, pages 267–288, 1996.